

Course Time: ONLINE

Course Location: ONLINE

Instructor: Scott Dickie (he/him/his)

Email Address: scott.dickie@anokaramsey.edu
** please include "Math 2230" in the subject line of all emails

Office: H107 (Coon Rapids campus)

Google Voice: 612-562-8450 (612-5MATH50)
This number rings to my cell phone, my office phone, or goes to my voicemail, depending on where I am. PLEASE LEAVE A MESSAGE if I do not answer.

Office Phone: 763-433-1749

Office Hours: TBD

Text: *Probability and Statistics for Engineering and the Sciences, 9th Edition, Devore*

New or used textbook is OK. You must have a *WebAssign* access code for this class. The text is available in the bookstore bundled with a WebAssign access code, or you may purchase a stand-alone *Cengage Unlimited* subscription, which provides access to an online version of the text. If you are unsure about your textbook purchase, please contact me before buying.

You need ready access to the internet. High-speed access is preferable. Check with your instructor if you're not sure your computer hardware/software will be sufficient.

Class Website: All our class materials, assignments, quizzes, grades, and communication outside the classroom will take place on our class D2L site, and on the WebAssign website, in addition to your student email account.

WebAssign: WebAssign is a homework manager that we will be using for all homework assignments and weekly quizzes in this course. All students must purchase a WebAssign access code (there is about a two-week grace period in which you can use WebAssign before you purchase the access code). To enroll in WebAssign, please follow the instructions posted on our class D2L site.

Student E-mail Account: I will be sending out all official announcements to your official school email address, so it is very important that you check your Anoka-Ramsey email on a regular basis (preferably daily). You can find information about your school email by following this link:

<http://www.anokatech.edu/TechnologyServices/Email.aspx?tab=2>

If you prefer to use your personal email, or maybe your email from another school, you will need to set up your ARCC email to forward to your personal email account (or wherever). To do this, please find the information at the link above, or contact the IT Helpdesk at 763-433-1510. You may still send me email from any account you would like, and I will reply to that same email account (unless you request otherwise).

You will be responsible for knowing all information that I send to your ARCC email account. Please do what you can to make sure you receive that information.

Required Materials: You will need a notebook and a graphing calculator. I recommend that you use a Texas Instruments TI-83 Plus or TI-84 Plus graphing calculator. You may use a different calculator if you like, however I may not be able to help you learn how to use it. You may not use communication-capable devices that might have a calculator feature (e.g., cell phone, iPod, etc.). Using your calculator to store notes is prohibited. Using additional programs on your calculator is prohibited, except when specified by me.

Add / Drop / Withdraw: Full refund is available until 6/5/26. The last day to add or drop this course is 6/5/26. The last day to withdraw from this course is 7/14/25.

Diversity Statement:

I am committed to honoring and respecting diversity in my classroom. A diverse world is an interesting world. It is my goal to make every student comfortable in my class, and with me as an instructor and helper on their way to accomplishing their academic goals.

Accommodations: Any student that feels they may need accommodations for a disability should contact me as soon as possible. Every student should have the opportunity to start from a level playing field. If you're not sure if accommodations would be appropriate for you, please see me or contact the Office for Students with Disabilities at 763-433-1350.

Active Military Members: Military members that are currently serving should keep me advised of all regularly scheduled military training and duties that conflict with scheduled course requirements. I will work with the student to address issues that arise. For further information on this, refer to MnSCU Procedure 5.12.1 Military Service and Disabled Veterans at <http://www.minnstate.edu/board/procedure/512p1.html> . You may contact the ARCC Veterans Services office at 763-433-1313.

Course Description: This is a calculus-based course in the study of probability and statistics. Topics include descriptive statistics, general probability, probability distributions, sampling, descriptive statistics, normal and binomial distributions, interval estimation and hypothesis testing, chi-squared and t-tests, two-sample inference, analysis of variance, correlation and linear regression analysis. Students will also learn how to use a statistical software package. Concepts are introduced at more advanced level than in MATH 1114.

Prerequisite: The course requires any one of these prerequisites:

- MATH 1401 with a grade of C or better, or equivalent.

Major Areas of Course Content:

1. Descriptive statistics
2. Probability and counting principles
3. Mathematical expectation
4. Discrete probability distributions
5. Density functions and continuous probability distributions
6. Sampling distributions and the Central Limit Theorem
7. Estimation and confidence intervals
8. Correlation and regression
9. Hypothesis testing
10. Analysis of variance

Minnesota Transfer Curriculum: Goal 04 - Mathematical/Logical Reasoning

- A. Illustrate historical and contemporary applications of mathematical/logical systems.
- B. Clearly express mathematical/logical ideas in writing.
- C. Explain what constitutes a valid mathematical/logical argument(proof).
- D. Apply higher-order problem-solving and/or modeling strategies.

Learner Outcomes: By the end of this course, a successful student will be able to do the following.

1. Calculate and analyze measures of central tendency and variability: mean, median, mode, range, variance, standard deviation, percentiles, and z-scores.
2. Construct and interpret visual summaries of data: frequency tables, histograms, box plots, and stem-and-leaf plots.
3. Understand basic concepts in probability, including combinatorics, independence, complementary events, conditional probability, and Bayes' theorem.
4. Calculate probabilities using the addition and multiplication rules, the fundamental counting principle, permutations, combinations, and sample spaces.
5. Understand the concepts of random variables and probability distributions; construct a probability distribution for a random variable; distinguish between discrete and continuous probability distributions.
6. Understand the concept of mathematical expectation; calculate mean, variance and covariance of random variables.
7. Calculate and interpret expected values, variances, and probabilities for discrete probability distributions, such as binomial, geometric, negative binomial, hypergeometric, and Poisson, with an emphasis on binomial and Poisson.
8. Identify density functions and use methods from calculus to determine mean, variance, and probabilities of density functions.
9. Calculate and interpret expected values, variances, and probabilities for continuous probability distributions, such as uniform, gamma, exponential, chi-squared, with an emphasis on normal distributions.
10. Apply the Central Limit Theorem to describe the sampling distributions and calculate probabilities.
11. Construct and interpret confidence intervals of population means, proportions, and variances.
12. Create, perform, and interpret hypothesis tests of population means, proportions, and variances.
13. Perform chi-square tests of goodness of fit, independence, and homogeneity.
14. Interpret scatter plots as describing form, direction and strength of relationships in bi-variate data, and compute and interpret the correlation coefficient for linear bi-variate data.
15. Use linear regression to investigate correlation of paired data and to make predictions.
16. Perform analysis of variance.
17. Use statistical software to find descriptive statistics for a data set, display the data, find probabilities, perform basic statistical inference, and perform linear correlation and regression analysis.

Lectures and Readings: You are encouraged to read each assigned section in the textbook *before* that section is covered in class (see course calendar). This will help you identify questions you might have, and any areas you have difficulty understanding. Lecture recordings will be made available to you on our class D2L site. You are encouraged to view all of the provided recordings *at least* once. Your viewing should be focused on understanding more than on mere completion. Lecture notes will be provided on the class D2L site. I recommend that you print them out ahead of time and bring them to class to facilitate your own notetaking.

Homework: Homework assignments will be completed in WebAssign, and will be graded by the computer. The problems selected represent a general cross-section of what you are expected to know, however they are not intended to be exhaustive (i.e. just because a problem wasn't "assigned" does not mean that you aren't expected to know how to do it). Homework assignments will typically be due by midnight on the Thursday or Sunday following the day they were assigned. Specific dates will be listed on WebAssign and on the course calendar. You may work on each homework assignment as much as you would like. Homework will be accepted past the due date with NO penalty, until 7/22/25. **I expect you to use the homework assignments appropriately to make sure that you understand the concepts and can perform the skills covered in this course.** *Note: You may use your textbook, as well as any of your notes, on homework. You may work with others on your homework, though you are expected to fully understand any answer you submit.*

Quizzes: Weekly quizzes will be completed in WebAssign, and will be graded by the computer. These quizzes will consist of problems (or slight variations thereof) from the assigned homework. Quizzes will typically be due before midnight on Thursday or Sunday nights. Specific dates will be listed on WebAssign and on the class calendar. You may submit each quiz up to 2 times, with your most recent submission counting toward your grade. For any submission beyond the first, you only need to re-do the parts you got wrong. Late quizzes will not be accepted. However, you do get to drop your lowest quiz score at the end of the semester, so if you miss one, it will not sink you. Just do not miss more than one! *Note: You may use your textbook, as well as any of your notes, on quizzes. You may not work with others on your quizzes.*

Tests: There will be four proctored tests given during the semester, plus an optional final exam. Check the course calendar for the dates of the tests and the sections that each will cover. If there are changes in the scheduling of tests, I will notify you well in advance. You will be given a selection of times to take your test, so you may work around any scheduling difficulties you have. There will also be options to take your tests at the ARCC Testing Center, if none of the offered time slots work for you. Scheduling with the Testing Center will need to be approved by the instructor and arranged beforehand. Tests will be taken online, and may include a take-home portion to be submitted via D2L. Except in *extreme* circumstances, missed tests cannot be made up. *Note: You will NOT be allowed to use your book, or any of your own notes, on any of the tests. However, I will provide you with some helpful notes, formulas, etc. for each test. You will be expected to bring your graphing calculator,*

pens/pencils, and some blank scratch paper (if you want it). I reserve the right to clear your calculator's memory before any test. Communication devices such as cell phones are not allowed to be out during tests.

Missed Test Policy: If you miss an exam, there is good news and bad news. The bad news is you cannot make-up the test. The good news is that at the end of the semester, I will replace your lowest test score with the optional final exam (unless your final is lower than all your test scores). So if your car won't start or your kids are sick, you don't have to worry too much about the exam and can concentrate on taking care of the problem (the car, the kids, or whatever). This policy applies to everyone, even if you do not actually miss a test. This means that if you have a bad day or just aren't as prepared as you should be for an exam, you have a second chance to show that you know the material by the time you take the final. (Warning: this policy will affect only ONE test grade so do not miss two or more tests).

Grading: Your grade will be based on your homework average (20% of your grade), your quiz average (20%), and your test average (60%). Letter grades will be assigned approximately as follows:

A 90% B 80% C 70% D 60% F below 60%

Attendance Expectations: I do not take attendance in class, but you are expected to demonstrate engagement in the class. Engagement may consist of classroom attendance, completion of online assignments, etc. It is ultimately up to my discretion whether or not you are meeting this expectation. **Failure to engage in the coursework for 10 consecutive school days could result in your removal from the course, without financial recompense (per ARCC policy).**

Getting Extra Help: If you find that you need some extra help, and you can't make it in to my office hours, I encourage you to use some of the other resources available here at the college. You can find information on our class website.

Cheating: I am against it, and will not tolerate it. Any instances of cheating will result in a zero for that assessment and a referral to the academic dean. Additional penalties will apply at my discretion.

HOW TO SUCCEED IN THIS CLASS:

1. Keep up with the schedule.
2. Attend all classes.
3. Do the homework.
4. Complete the quizzes on time.
5. Study for the tests.
6. Don't miss any tests.
7. IF YOU NEED HELP, GET IT SOONER RATHER THAN LATER!

The instructor reserves the right to make changes or additions to this syllabus. You will be notified of any changes by email and by notification on the course website.



Be Respectful. Be Responsible. Rise to the Challenge.

By enrolling in this course, you have become a welcome member of a community of learners, an honor and a privilege for each one of us. Your membership in this community and your success in this course are dependent upon your ability to do the following:

Be Respectful. Respect your instructor as an expert in his or her subject area and as the person responsible for facilitating a productive course for everyone. Respect each of your classmate's right to a valuable class experience, free of offensive language, intolerance, or harassment of any kind. Respect these facilities and our time together by eliminating all distractions, especially cell phones, iPods, and other gadgets, and by refraining from disruptions of any kind, including sleeping in class or talking when no formal class discussion is taking place. Finally, respect yourself by participating fully in each class session and making the most of this learning opportunity.

Be Responsible. As a student in this college course, you are entirely responsible for your own success. You are responsible for reading and following the syllabus. It's expected that you arrive to each class session on time, with assigned work completed, ready to participate fully. If you miss class, you are responsible for the consequences. You are also responsible for obtaining notes, hand-outs, assignments, and syllabus adjustments. Finally, you are responsible for being an active participant in this class rather than a passive observer.

Rise to the Challenge. College-level courses are demanding. They require deeper thinking, more effective writing, and greater personal involvement than many students realize. In order to succeed at this level, you must be willing to accept the challenges presented by the course material, your instructors, and a rigorous schedule. One of the rewards of this challenge can be the discovery that you are capable of much more than you imagined. Therefore, expect great things from yourself, work hard to achieve them, and seek help when you need it. The other members of this community of learners are here to support you, but it's up to you to rise to the challenge.